# ARYAN GUPTA

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## EDUCATION

**Imperial College London** Sep 2026 - Sep 2027
*MSc Applied Computational Science & Engineering - Incoming*

**University of Michigan** Sep 2019 - May 2023
*BSE Engineering Physics, cum laude | Minors: Electrical Engineering, Computer Science, Entrepreneurship*

## PROFESSIONAL EXPERIENCE

**Technip Energies - Houston, TX** Aug 2023 - Jul 2026

**Technology Developer I, Applied Physics / Carbon-Capture R&D** *(Aug 2023-Jul 2025; Feb-Mar 2026)* - Built the team's first-principles MD/DFT capability across HPC and cloud; linked atomistic outputs into diffusion/CFD models and experimental workflows.

- Reduced simulation wall time by roughly two-thirds through GPU/cloud and parallel-I/O workflows; automated parameter sweeps and terabyte-scale image analysis, cutting multi-day manual analysis to hours.
- Developed uncertainty-aware particle analysis, surrogate models, active-learning/optimization workflows, and verification tests; authored 15 reviewed technical notes and a 70-page market/business analysis used in funding and roadmap decisions.

**Americas AI Solutions Initiative - Co-Founder, Co-Lead & Technical Lead** *(Dec 2025-Jul 2026)* - Co-led an internal initiative spanning 4 operating centers and 8 engineering functions while remaining in an engineering role.

- Scoped and led engineering AI workflows across structured data and technical documents, including natural-language-to-SQL and vendor-document analysis; one bid-review workflow reduced manual review effort by about 80%.

**Specialist Engineer I - Instrumentation & Controls** *(Jul 2025-Jul 2026)* - Worked on hydrogen, carbon-capture, and brownfield scopes; reviewed instrumentation/control documentation against P&IDs, PFDs, Cause & Effect diagrams, narratives, and standards.

## SELECTED RESEARCH & OPEN SOFTWARE

**WaveSleuth-Devito: Inverse Wave Playground v1.0** Jun-Aug 2026
*Devito-backed 2D acoustic inverse-problem sandbox | GitHub: guptaaryanr/wavesleuth-devito*

- Built deterministic forward simulation, sparse acquisition, circle/ellipse/coarse-mask inversion baselines, blind challenges, uncertainty summaries, active sensing, scoring, and release validation.
- Canonical v1.0 toy challenges reached IoU 1.000 (mask), 0.895 (circle radius+velocity), 0.800 (circle), and 0.787 (ellipse); deliberately scoped as an inspectable sandbox rather than a field-scale inversion claim.

**Trajectory-Only Structural Regime Detection** Feb-May 2026
*Preprint DOI: 10.5281/zenodo.20278240 | Code DOI: 10.5281/zenodo.19465797*

- Developed a trajectory-only detector for structural changes in deterministic nonlinear systems, keeping model-aware transition definitions separate for post-hoc evaluation.
- Tested on FitzHugh-Nagumo and forced van der Pol systems with boundary extraction, lead-distance, nuisance, specificity, robustness, and ablation studies.

**ThermoBench-Consist v1.0** Sep-Oct 2025
*Preprint DOI: 10.5281/zenodo.17489426 | Code DOI: 10.5281/zenodo.17330440*

- Built a CPU-only diagnostic for ML thermodynamic surrogates using density-pressure monotonicity, positive compressibility, Clapeyron consistency, and speed-of-sound checks, with guardrails for fragile thermodynamic regions.

**Assessing the Limits of GNNs for Vapor-Liquid Equilibrium** May-Sep 2025

*arXiv:2509.10565*

Negative-results study showing that acceptable pointwise prediction errors can still produce derivative pathologies and liquid-phase inconsistencies that make learned surrogates unusable inside VLE solvers.

**Energy-Efficient Robotics Software: Systematic Literature Review** Jan-Aug 2025

*arXiv:2508.12170 | Replication DOI: 10.5281/zenodo.16907564*

Synthesized 79 post-2020 studies, proposed a taxonomy of energy-aware techniques and a reporting checklist, and released the screening/figure-generation replication package.

**REB-1: Robot Energy Benchmark v0.1** Aug 2025

*Code DOI: 10.5281/zenodo.17204853*

Built a lightweight CLI/notebook benchmark for reproducible power traces, Wh integration, and user-specified grid-intensity conversion to gCO2 for quick autonomy-workload A/B tests.

## EARLIER RESEARCH, TEACHING & BUILDING

**BIRDS Lab, University of Michigan - Research Assistant** *(2021-2023)* - Built and debugged adaptable quadcopter hardware/software across ESP32/FTDI telemetry, power electronics, flight-control software, and mechanical integration.

**EECS 314 Circuit Design & Analysis - Instructional Aide** *(2022-2023)* - Led discussion groups for about 80 students and a 20-student circuits lab; supported course delivery for 160+ students.

**MiTEE - Plasma Team / Structures Liaison** *(2020)* - Worked on electrodynamic-tether propulsion feasibility for pico/nanosatellites, including thermodynamic analysis, electronics prototyping, and subsystem interfaces.

**HelixCases - Co-Founder & Software Developer** *(2022)* - Co-built a university technical-recruiting competition platform reaching 550+ students at 20 universities; secured a $10k Lyft contract.

## SELECTED ENGINEERING PROJECTS

**Electric Longboard Remote Controller** *(2022)* - STM32L010C6T6 remote with UART/LPUART phone link, battery/display feedback, modified VESC firmware, and a 3D-printed enclosure.

**Out-of-Order Processor Architecture** *(2022)* - MIPS R10K-style SystemVerilog core with 2-way superscalar issue, I/D caches, LSQ, branch prediction, precise exceptions, baseline SMT, and a visual debugger.

**Low-Autonomy Follower Robot** *(2022)* - Two-Zumo leader/follower system using Nucleo boards, LED-encoded commands, PixyCam vision, ultrasonic ranging, and PID control.

## LEADERSHIP & SERVICE

**Project RISHI - Fundraising Co-Chair** *(2020-2021)* - Helped secure $16k toward a $35k budget through grant applications and campus fundraising.

**StartUM - Member & Mentor** *(2021-2022)* - Mentored an early-stage medical-volunteer startup through validation and a change in target market.

**Citizens for Animal Protection - Volunteer** *(2023-2024)* - Dog walking, kennel cleaning, and day-to-day shelter support in Houston.

## SKILLS

**Scientific computing & simulation:** Python, C/C++, MATLAB/Simulink, NumPy, SciPy, LAMMPS, NWChem, Devito, MD/DFT, ODE/PDE simulation, thermodynamics, UQ

**ML, optimization & evaluation:** PyTorch, scikit-learn, OpenCV, surrogate modeling, active learning, numerical optimization, benchmarking, validation/evaluation harnesses

**Systems & tooling:** Linux/Unix, Git, Docker, CI/GitHub Actions, SLURM, GCP, SQL/PostgreSQL, OCR/PDF parsing, embedded STM32/ARM, ESP32, UART, FreeRTOS, PID control

## HONORS & CREDENTIALS

Six Sigma Green Belt (CSSGB), Technip Energies (2026) | Foundational Finance, University of Michigan (2020) | Cum Laude, University of Michigan (2023) | University Honors & Dean's List (2019, 2020)